Build a complete prediction

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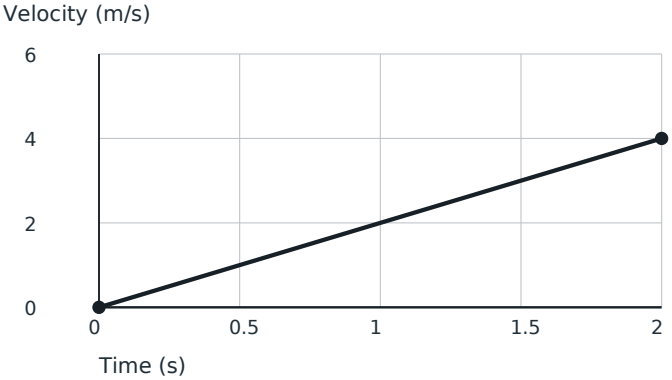
Learning target: Connect net force, velocity change, and final position in one model.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 17.*
At rest: friction 20 N opposite the push. Sliding: net force 20 N with the push.

02. Worked example: explain the model
Triangle base 2 s, height 4 m/s, area 4 m; net force arrow right.

Model / diagram check



03. Start with forces
Net 4 N right; a 2 m/s² right.

04. Predict motion
vf 6 m/s right; average 3 m/s right;Δx 9 m right; xf 10 m.

05. Explain a limit
Changing push/friction, etc. More opposing friction reduces net force and acceleration for fixed mass.

06. Review: connect an earlier idea *Review: Unit 1, Day 8.*
a=2.8 m/s² east; average velocity 7 m/s east; displacement 35 m east.

07. Exit check
a 3 m/s² right; vf 6 m/s right; constant force/mass. Assessment looks at justified method and assumption.

08. STRETCH: test the model
Same acceleration 2 m/s², mass 4 kg, net force 8 N.

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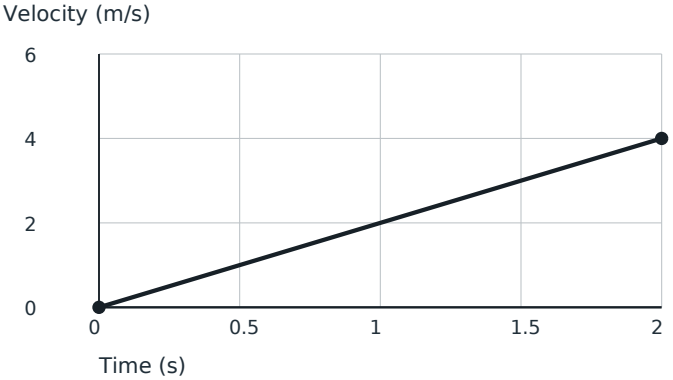
Learning target: Connect net force, velocity change, and final position in one model.

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01. Warm-up: bring back an earlier idea *Review: Unit 1, Day 17.*
N=29.4 N; static limit 14.7 N. Low push: friction 6 N, a=0. High push: kinetic friction 11.76 N, a=2.08 m/s².

02. Worked example: explain the model
Triangle base 2 s, height 4 m/s, area 4 m; net force arrow right.

Model / diagram check



03. End point versus displacement
Endpoint (-20, 60) km; displacement (80, 60) km, magnitude 100 km; separation√4000=63.2 km.

04. Add bounds
δx=1+20(0.1)=3 km→[-23,-17];δy=2 km→[58, 62].

05. A new force
a=0.2 m/s²; extra Δv=2 m/s; extra Δx=10 m.

06. Review: connect an earlier idea *Review: Unit 1, Day 8.*
Stops after 7 s at distance 122.5 m. Extending to 9 s predicts -10 m/s, which assumes continued acceleration after stopping and reversal.

07. Exit check
Should report (-20, 60) km with x[-23,-17], y[58, 62]; limitations gravity/Earth motion/3 D, etc. A bound is not a probability.

08. STRETCH: test the model
a=3 m/s², v_f=10 m/s, x_f=3-8+24=19 m. Turns at 2/3 s; cannot use speed as a signed velocity.